# ZOO202 Animal Diversity Chordates

## FINAL TERM PAPER PREPARATION BEST FILE BY

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## BS-ZOOLOGY 4TH SEMESTER

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**Reproduction in tunicates :**

Tunicates are generally hermaphrodites that sexually reproduce by cross-fertilization. That means that an individual does not use its own sperm to fertilize its eggs. Instead, they obtain sperm released into the open ocean by other individuals. Fertilization is usually internal.

**Nitrogenous excretion in fishes:**

Fishes in water excrete ammonia as the major nitrogenous waste through gills, but gills of air-breathing fishes are modified for air-breathing or largely replaced by air-breathing organs.

**spectacles definition:**

Lenses set in a frame that holds them in front of the eyes, used to correct errors of refraction or to protect the eyes.

**Examples of annilids:**

1= Earthworm. 2=Leeches. 3= Lugworms. 4= Polychaetes.

**What is preening?**

Preening is a maintenance behaviour found in birds that involves the use of the beak to position feathers, interlock feather barbules that have become separated, clean plumage, and keep ectoparasites in check.

**Define pheromones:**

A pheromone is a chemical that an animal produces which changes the behavior of another animal of the same species. Some describe pheromones as behavior-altering agents.

**define cirri:**

a cirri is a long, thin structure in an animal similar to a tentacle but generally lacking the tentacle's strength, flexibility, thickness, and sensitivity.

**Arrector pili muscle:**

A band of smooth muscle, called the arrector pili muscle. Arrector pili muscles are under the control of the autonomic nervous system, which also controls a mammal's "fight-or-flight" response.This is a tiny muscle that attaches to the base of a hair follicle at one end and to dermal tissue on the other end. In order to generate heat when the body is cold, the arrector pili muscles contract all at once, causing the hair to "stand up straight" on the skin.

**Convergent evolution:**

Convergent evolution is when different organisms independently evolve similar traits. For example, sharks and dolphins look relatively similar despite being entirely unrelated.

**Define amphibians:**

Amphibians are small vertebrates that need water, or a moist environment, to survive. The species in this group include frogs, toads, salamanders, and newts.

**Difference between monoestrus, diestrus, and polyestrus:**

Monoestrous animals are those which have only one oestrous cycle in a year and hence breed only once a year.

Diestrus is a period of sexual quiescence that intervenes between two periods of estrus.

polyestrus having several estrus cycles annually or during a breeding season.

**Altricial and precocial:**

Altricial are being hatched or born or having young that are hatched or born in a very immature and helpless condition so as to require care for some time.

Precocial: (young bird or other animal) hatched or born in an advanced state and able to feed itself almost immediately.

## 4 unique charateristics of chordates:

At developmental stages:

1= Presence of notochord 2= Presence of nervechord 3= Presence of post-annel tail 4= Presence of phyrengial slites

## Role of salivary glands in lamprey:

When lampreys attack host fishes to suck blood and flesh, their buccal glands could secrete enough proteins to suppress blood coagulation, nociception, oxidative stress, immune response, as well as other adverse effects encountered during their parasitic lives.

## Give an example of urochordates?

Tunicates or sea squirts.

## What is Asetivation?

Aestivation (Latin: aestas (summer); also spelled estivation in American English) is a state of animal dormancy, similar to hibernation, although taking place in the summer rather than the winter.

## What is hibernacula?

Individuals that are usually solitary may migrate to a common site to spend the winter. Heat loss from these groups, called hibernacula.

## What is hibernation?

Hibernation is a period of winter inactivity in which the hypothalamus of the brain slows the metabolic, heart, and respiratory rates. In preparation for hibernation,mammals usually accumulate large quantities of body fat.

## What is PLUMAGE?

The covering of feathers on a bird is called the plumage.

## what is annuli?

Earthworms' bodies are made up of ring-like segments called annuli. These segments are covered in setae.

## What is ecdysis?

The process of shedding the old skin (in reptiles) or casting off the outer cuticle (in insects and other arthropods).

## Homodont and heterodont:

| Homodont | Heterodont: |
|---|---|
| In reptiles, the teeth are uniformly conical, a condition referred to as homodont | In mammals, the teeth are often specialized for different functions, a condition called heterodont. |

## Types of flight?

**1= Gliding flight**: during gliding flight the wing is stationary, and a bird loses altitude. Waterfowl coming in for a landing use gliding flight

**2= Flapping flight:** generates the power for flight and is the most common type of flying. Many variations in wing shape and flapping patterns result in species-specific speed and maneuverability

**3= Soaring flight** allows some birds to remain airborne with little energy expenditure. During soaring, wings are essentially stationary, and the bird utilizes updrafts and air currents to gain altitude. Hawks, vultures, and other soaring birds are frequently observed circling along mountain valleys, soaring down-wind to pick up speed and then turning upwind to gain altitude.

**4= Hovering flight**. They hover in still air by fanning their wings back and forth (50 to 80 beats per second) to remain suspended in front of a flower or feeding station. Eg Hummingbirds.

## Shivering and non shivering?

| Shivering | Non-shivering |
|---|---|
| **Shivering thermogenesis** is muscular activity that generates large amounts of heat but little movement. | **Nonshivering thermogenesis** involves heat production by general cellular metabolism and the metabolism of special fat deposits called brown fat. |

## Difference bw ventricles and proventriculus?

| Ventricles | Proventriculus |
|---|---|
| The ventriculus, or muscular stomach (also known as the "gizzard"), contains two opposing sets of muscles used for grinding food. | The proventriculus, or glandular stomach, is continuous with the esophagus at the level of the base of the heart and contains digestive (pepsinogen-secreting) and mucous glands. |

## Define hair?

Hair is a keratinized derivative of the epidermis of the skin and is uniquely mammalian. It is seated in an invagination of the epidermis, called a hair follicle. The hair is a slender threadlike outgrowth of the epidermis of an animal.

## Example of anurans?

any of an order (Anura) of amphibians comprising the frogs, toads, and tree frogs all of which lack a tail in the adult stage and have long hind limbs often suited to leaping and swimming.

## What is order of humans?

the order of humans is primates.

## Binocular vision?

vision using two eyes with overlapping fields of view, allowing good perception of depth.

## PINEAL body?

A pineal body on the roof of the forebrain appears to stimulate ovarian development and to regulate other functions influenced by light and dark periods**.**

## Define Spectacle?

In snakes and some lizards, the upper and lower eyelids fuse in the embryo to form a protective window of clear skin, called the spectacle. (During ecdysis, the outer layers of the spectacle become clouded and impair the vision of snakes.

## Arrector pili muscles

A band of smooth muscle, called the arrector pili muscle, runs between the hair follicle and the lower epidermis. Arrector pili muscles are under the control of the autonomic nervous system, which also controls a mammal's "fight-or-flight" response.

## Counter current

Countercurrent heat-exchange systems may help regulate heat loss from exposed areas. Arteries passing peripherally through the core of an appendage are surrounded by veins that carry blood back toward the body. When blood returns to the body through these veins, heat transfers from arterial blood to venous blood and returns to the body rather than being lost to the environment . When excess heat is produced, blood is shunted away from the countercurrent veins toward peripheral vessels, and excess heat is radiated to the environment.

## Negative pressure mechanisms?

In most reptiles, a negative-pressure mechanism is responsible for lung ventilation. A posterior movement of the ribs and the body wall expands the body cavity, decreasing pressure in the lungs and drawing air into the lungs. Air is expelled by elastic recoil of the lungs and forward movements of the ribs and body wall, which compress the lungs. The ribs of turtles are a part of their shell; thus, movements of the body wall to which the ribs attach are impossible. Turtles exhale by contracting muscles that force the viscera upward, compressing the lungs. They inhale by contracting muscles that increase the volume of the visceral cavity, creating negative pressure to draw air into the **lungs.**

## Types of mammalian teeth

1= **Incisors** are the most anterior teeth in the jaw. They are usually chisel-like and used for gnawing or nipping.

**2= Canines** are often long, stout, and conical, and are usually used for catching, killing, and tearing prey. Canines and incisors have single roots

**3= Premolars** are positioned next to canines, and have one or two roots and truncated surfaces for chewing.

**4=Molars** have broad chewing surfaces and two (upper molars) or three (lower molars) roots.

## Characterstics of mammals

1= Any member of the group of vertebrate animals in which the young are nourished with milk from special mammary glands of the mother. 2= In addition to these characteristic milk glands, mammals are distinguished by several other unique features. Hair is a typical mammalian feature, although in many whales it has disappeared except in the fetal stage. 3= A chain of three tiny bones transmits sound waves across the middle ear. 4= A muscular diaphragm separates the heart and the lungs from the abdominal cavity. Only the left aortic arch persists. (In birds the right aortic arch persists; in reptiles, amphibians, and fishes both arches are retained.) 5= Mature red blood cells (erythrocytes) in all mammals lack a nucleus; all other vertebrates have nucleated red blood cellsThe brain of mammals has an expanded cerebral cortex that processes information from various sensory structures. 6= Metanephric kidneys permit mammals to excrete urea without excessive water loss. 7= Complex behavior patterns enhance mammalian survival 8= Most mammals are viviparous and have reproductive cycles that help ensure internal fertilization and successful development.

**What is pygostyle?**

A plate of bone that forms the posterior end of the vertebral column in most birds and is formed by the union of vertebrae.The posterior caudal vertebrae are fused into a pygostyle, which helps support the tail feathers that are important in steering.

**Define migration**

seasonal movement of animals from one region to another.

**Orders of reptile**

Class Reptilia is made up of four orders: Squamata, Testudines, Crocodilia and Rhynchocephalia.

**Characterstics of chordate**

1= Bilaterally symmetrical 2= deuterostomateanimals 3= Four unique characteristics present at some stage in development: notochord, pharyngeal slits or pouches, dorsal tubular nerve cord, and postanal tail Presence of an endostyle or thyroid gland 4= Complete digestive tract.5= Ventral, contractileblood vessel (heart) 6= The four characteristics listed in number 2 are unique to chordates and are discussedfurther in the paragraphs that follow. 7= The phylumis named after the notochord 8= It consists of a connective-tissue sheath that encloses cells 9= This arrangement gives the notochord some turgidity, which prevents compressionalong the anteroposterior axis. 10=At the same time, the notochord is flexible enough to allow lateral bending 11= In most adult vertebrates, cartilage or bone partly or entirely replacesthe notochord.

**Name of opening tunicates?**

The adult tunicate body is shaped like a small barrel with two large openings called siphons**.**

**Inspiration and expiration**

**Inspiration** results from contraction of the diaphragm and expansion of the rib cage, both of which decrease the intrathoracic pressure and allow air to enter the lungs.

**Expiration** is normally by elastic recoil of the lungs and relaxation of inspiratory muscles, which decreases the volume of the thoracic cavity. The contraction of other thoracic and abdominal muscles can produce forceful exhalation.

**Give two major development that define gnathostimes?**

**Gnathostomes**

Two major developments in vertebrate evolution were the appearance of jaws and paired appendages.

**Define chameleon:** Chameleons are adapted to arboreal lifestyles and use a long, sticky tongue to capture insects.

**write three names of pelvic girdle of amphibians?**

The pelvic girdle, also known as the hip bone, is composed of three fused bones: the ilium, ischium and the pubic bone.

**write the name of ancient bird found in china?**

Gansus yumenensis is the first Mesozoic bird found in China.

Write difference between crocodile and alligator:

Alligators are dark green or black, have u-shaped snouts, tiny black spots and are smaller than crocs. Crocodiles are bigger, have v-shaped snouts and are more aggressive.

**Define clutch?**

The group of eggs laid and chicks produced by a female is called a clutch.

**Ram ventilation:** Ram ventilation is a simpler process in which a fish swims forward with its mouth open, taking in water that passes over the gills. The drawback to ram ventilation is that the fish has to swim continuously to be able to continue breathing.

**What is siphons?**

The adult tunicate body is shaped like a small barrel with two large openings called siphons.

**lateral line system:**

lateral line system, also called lateralis system, a system of tactile sense organs, unique to aquatic vertebrates from cyclostome fishes (lampreys and hagfish) to amphibians, that serves to detect movements and pressure changes in the surrounding water.

**Describe Types of layers on the reptilian egg:**

1= amnion 2= chorion 3= allantois

**Explain Mammary Gland in mammals?**

Mammary glands are functional in female mammals and are present, but nonfunctional in males. The milk that mammary glands secrete contains water, carbohydrates (especially the sugar lactose), fat, protein, minerals, and antibodies. Mammary glands are probably derived evolutionarily from apocrine glands and usually contain substantial fatty deposits. Monotremes have mammary glands that lack nipples. The glands discharge milk into depressions on the belly, where the young lap it up. In other mammals, mammary glands open via nipples or teats, and the young suckle for their nourishment.

**Stomach of birds:**

Birds have a two part stomach, a glandular portion known as the proventriculus and a muscular portion known as the gizzard. Hydrochloric acid, mucus and a digestive enzyme, pepsin, are secreted by specialized cells in the proventriculus and starts the process of breaking down the structure of the food material.